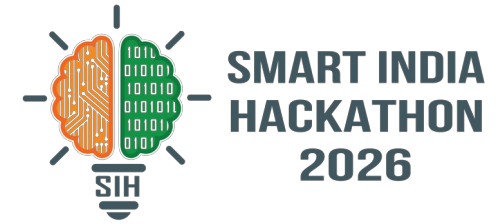


SMART INDIA HACKATHON 2026



TITLE PAGE

Problem Statement ID: SIH26183

Problem Statement Title

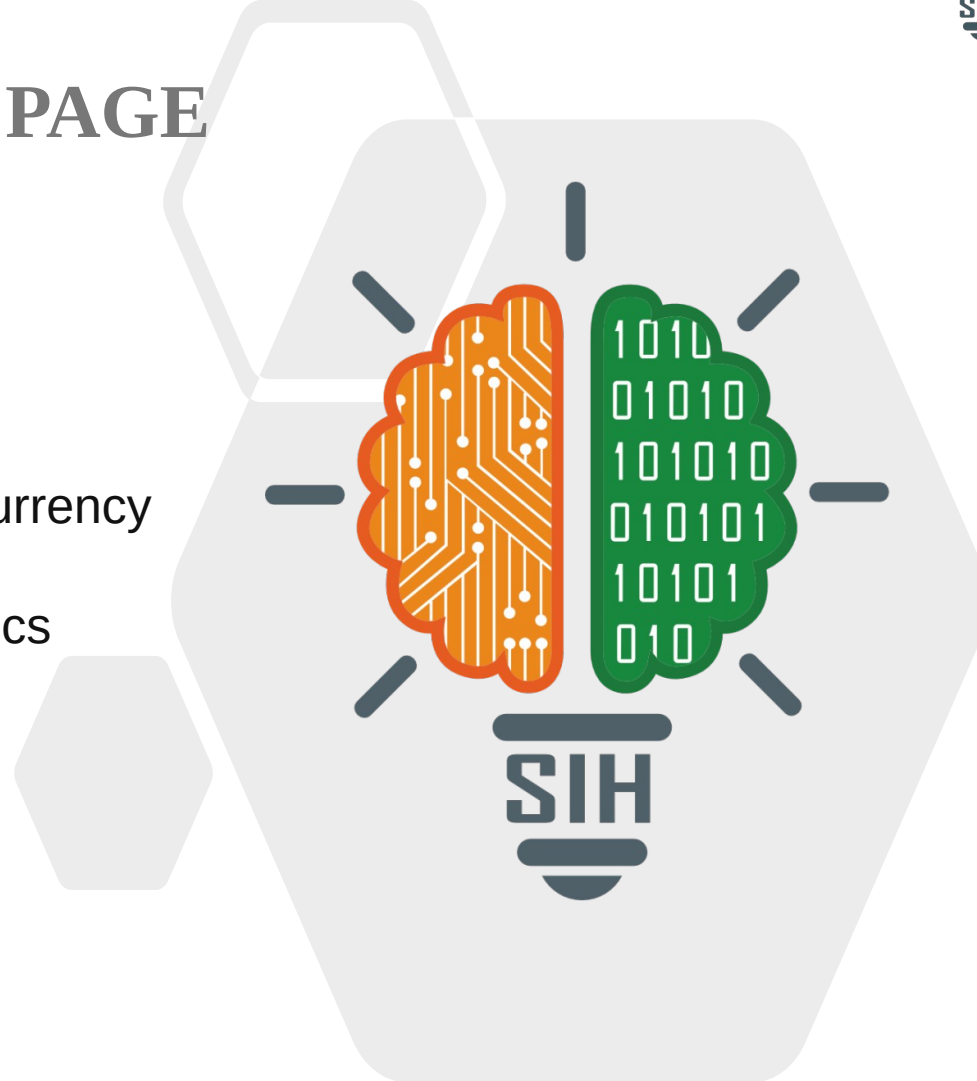
Real-Time Identification of Fraud-Linked Cryptocurrency Exchanges from Victim-Reported Suspect Wallet Addresses through Automated Blockchain Analytics

Theme: Blockchain & Cybersecurity

PS Category: Software

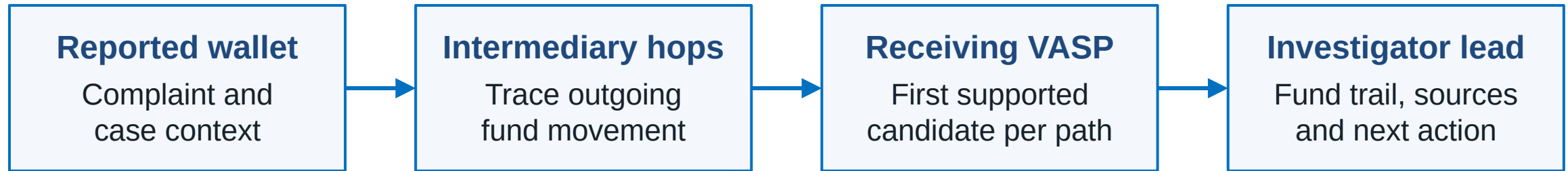
Team ID: 156766

**Team Name (Registered on portal):
COLD COFFEE**



Proposed Solution (Describe your Idea/Solution/Prototype)

Detailed explanation of the proposed solution



How it addresses the problem

Continuous monitoring will alert investigators when funds move or a receiving VASP appears.

A case report will support review, preservation requests and coordination with the VASP.

Innovation and uniqueness of the solution

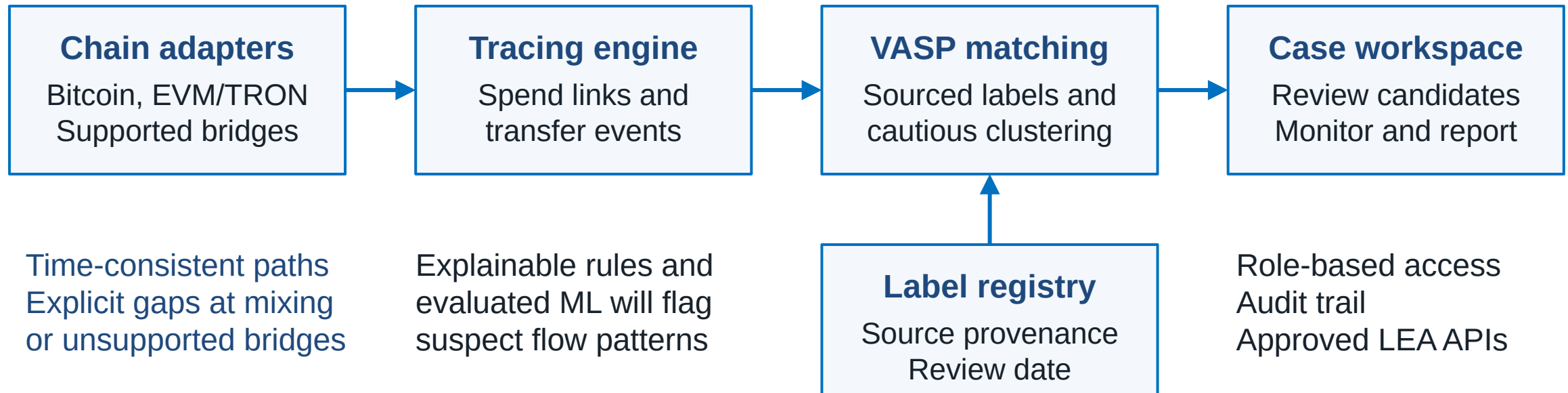
Each lead links the transaction path to attribution sources and an investigative next step.

New transactions and revised labels will update findings while preserving earlier analysis.

Technologies to be used

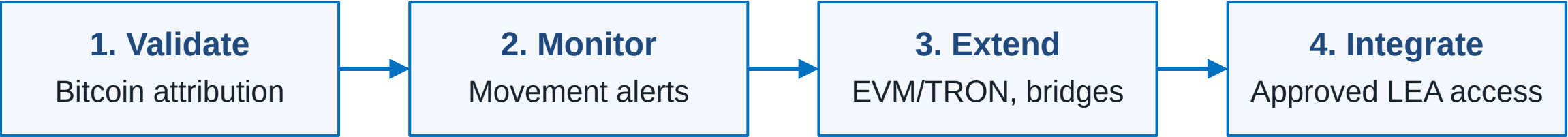
React dashboard and FastAPI services, queued workers, incremental transaction indexing, persistent case storage and a versioned VASP-label registry.

Methodology and process for implementation



Findings will distinguish observed transactions, inferred links and unresolved attribution.

Analysis of feasibility of the idea



Label supply: reviewed public disclosures and authorized VASP data, with source history.

Potential challenges and risks

Strategies for overcoming these challenges

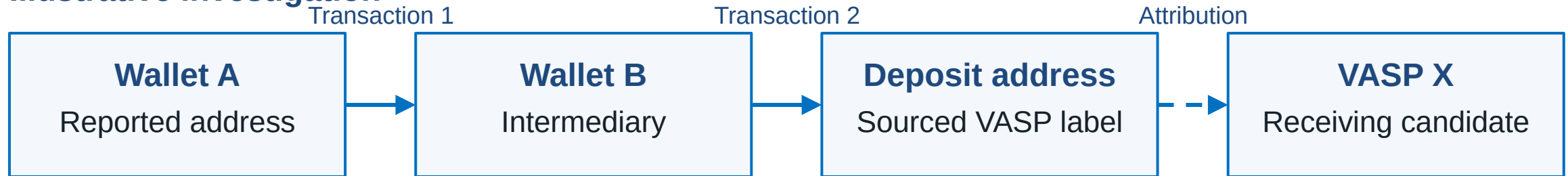
Uncertain labels and flows	Version label sources, review clusters and show unresolved paths.
Provider limits and large histories	Index incrementally, cache queries and bound graph traversal.
Sensitive cases and restricted access	Role-based permissions, audit logs and approved NCRP/SAHYOG APIs.

Validation: held-out labelled paths and hard negatives. Measure precision, coverage and alert latency.

Potential impact on the target audience

Investigators will receive a prioritized receiving-VASP lead with a reviewable fund trail.

Illustrative investigation



Finding Two-hop path, transaction IDs, amounts, timestamps and label source.

Next action Investigator reviews the lead and prepares a VASP preservation request.

Fictional illustration. Transaction paths alone do not prove exact fund continuity or wrongdoing.

Benefits of the solution (social, economic, environmental, etc.)

Social

Support timely intervention

Economic

Reduce repeated tracing work

Operational

Standardize VASP coordination

Details / Links of the reference and research work

Address clustering	Meiklejohn et al. (2013), A Fistful of Bitcoins: Characterizing Payments Among Men with No Names. ACM IMC paper
ML-assisted analysis	Weber et al. (2019), Anti-Money Laundering in Bitcoin: Experimenting with Graph Convolutional Networks for Financial Forensics. arXiv:1908.02591
Blockchain data	Blockstream Esplora: address and transaction API documentation. Official API documentation
Attribution sources	Binance, Our Commitment to Transparency (2022): published wallet disclosures. Historical disclosure; not a deposit-account registry
AML context	FATF (2021), Updated Guidance for a Risk-Based Approach to Virtual Assets and VASPs. FATF guidance
Evidence handling	NIST SP 800-86 (2006), Guide to Integrating Forensic Techniques into Incident Response. NIST guidance; not a guarantee of legal admissibility